

Nirma University
Institute of Technology
Semester End Examination (IR), May - 2026
M. Tech. in Computer Science and Engineering /
M. Tech. in Computer Science and Engineering (Data Science), Semester-II

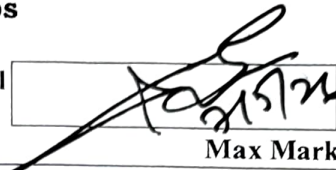
6CS374ME25 MLOps

Roll/

Exam No.

25MCD005

Supervisor's Initial
with date



Time : 3 Hours

Max Marks: 100

Instructions:

1. **All questions are compulsory. (No optional questions)**
2. **Use section-wise separate answer book.**
3. Figures to right indicate full marks.
4. Draw neat sketches whenever necessary.
5. Assume suitable data wherever necessary and state your assumptions clearly.

SECTION-I

Q-1 Answer the following questions.

[24]

A
CO3
BL4

Draw the architectural diagram of Kubernetes. Elaborate on core components of the Control Plane.

12

B
CO4
BL6

Analyze the "Continuous Integration" phase of a pipeline. Identify and describe the first four steps that occur immediately after a developer **commits** code, and explain why "Artifact Generation" serves as the bridge between CI and CD. Also, formulate a Declarative Jenkinsfile for a Python-based ML application. The pipeline must be restricted to a node with the label 'windows' and include the following five stages: Checkout, Linting (using flake8), Unit Testing (using pytest), Docker Build, and Cleanup. You must use bat or sh commands appropriately and include a failover mechanism for the testing stage to ensure the pipeline does not crash if no tests are found.

12

Q-2 Answer the following questions.

[16]

A
CO3
BL4

Compare the four primary types of Kubernetes Services. For each service type, provide a concrete example demonstrating its exact network routing flow from source to destination.

10

B
CO3
BL5

Explain the concept of "Quorum" in etcd and mathematically prove why production clusters strictly require an odd number of master nodes.

6

Q-3
CO4
BL3

Demonstrate the process of setting up a reproducible monitoring configuration for an ML pipeline using a Prometheus configuration file. Explain each section of this configuration file.

[10]

SECTION-II

Q-4 Answer the following questions.**[20]**
12**A**
CO4
BL6

Construct a Multi-Stage Dockerfile for a Python Flask application. In the first stage (build), use a full Python image to install heavy dependencies like scikit-learn and numpy to create a virtual environment. In the second stage (production), use a lightweight python-slim image and copy only the necessary artifacts from the first stage. Explain every line of your Dockerfile. In your explanation, justify why this multi-stage approach is superior to a single-stage build in terms of security and final image size for a scalable ML pipeline.

B
CO1
BL2

Distinguish between Centralized Version Control Systems (CVCS) and Distributed Version Control Systems (DVCS). In your explanation, identify the single point of failure associated with CVCS and explain how the distributed nature of Git provides a more reliable backup mechanism. Also, describe the three main sections of a Git project: the Working Directory, the Staging Area (Index), and the Local Repository. Trace the path of a file change as it moves through these three stages using the appropriate Git commands.

8

Q-5 Answer the following questions.**[18]**
12**A**
CO2
BL6

Write a Python script using `sklearn.pipeline.Pipeline` and `sklearn.model_selection.GridSearchCV` for a numerical dataset. Your code must: (i) Initialize a Pipeline that chains a scaler and a classifier. (ii) Define a dictionary for a Grid Search that tunes at least one parameter for the scaler and one for the classifier. (iii) Provide the commands to initialize the grid search with cross-validation, fit it on training data, and retrieve the best parameters. Also, describe the naming convention required in the parameter grid to target hyperparameters within specific pipeline stages. Justify why tuning the entire pipeline is superior to tuning the model in isolation.

B
CO2
BL3

(i) A developer needs to move a specific model-training image to a secondary server while keeping all metadata and layer history intact. Demonstrate the sequence of two commands required to save this image to a file and then load it on the new host. (ii) Contrast this with a scenario where you only need a filesystem snapshot of a running container for a quick backup. Identify the specific command used to create this flattened snapshot and explain why it is less suitable for reproducible pipelines than the method in part (i).

6

Q-6
CO4
BL5

Distinguish between an Application Server (such as Waitress, Gunicorn, or uWSGI) and a Web Server (such as Nginx or Apache). Justify the industry best practice of pairing an application server with a reverse proxy like Nginx, highlighting at least three benefits. Explain the significance of the WSGI (Web Server Gateway Interface) standard for Flask applications. Describe the specific networking implications of binding a server to 0.0.0.0 versus a specific IP address, and explain why 0.0.0.0 is a common requirement for containerized environments.

[12]